Week 7: Take-away games

Adult guide built from the problem files · Unpiloted.

The answers, hints and routes are in the adult guide, week-07-facilitator.pdf, whose source is guide-src/facilitator.tex. First route = the guide's core problems.

Problems and routes

Problem	K–1	2–3	4–5	route
Piles of 2 to 8: first or second?	P1			first
Play the grown-up from 10	P2			first
Play your partner from 10, taking turns going first	P3			first
The first take that wins from 5, 7, 8 and 10	P4			first
Your partner chooses the start, you choose who goes first	P5			first
Moves of 1, 2 or 3 from 10	P6			return
Moves of 1 or 4 from 8	P7			return
Two piles, taking 1 or 2 from one pile	P8			return
Last counter loses: piles of 1 to 6	P9			return
Piles of 2 to 8 with moves of 1, 3 or 4		P1		first
Squares to leave on 0 to 20, moves of 1, 3 or 4		P2		first
Play the adult from a start of 10 to 20		P3		first
Lena's claim from 9		P4		first
Omar takes as many as he can, from 13		P5		first
Piles of 50 and 100		P6		return
Two piles: 5 and 5, 5 and 6		P7		return
Piles with moves of 1 or 4		P8		return
Squares to leave for 1 or 4, and for 1, 3 or 5		P9		return
Piles with moves of 1 or 2			P1	first
Squares to leave for 1 or 2, and for 1, 2 or 3			P2	first
Piles with moves of 1, 3 or 4			P3	first
Squares to leave on 0 to 30, moves of 1, 3 or 4			P4	first
Kai's greedy first move			P5	first
A pile of 100 under three rules			P6	first
Squares to leave for 1 or 4, 2 or 3, and 1, 3 or 5			P7	return
Does the 1, 3, 4 pattern go on forever?			P8	return
Choose moves that give 0, 5, 10 or 0, 3, 6			P9	return
Last counter loses: piles with moves of 1 or 2			P10	return
Last counter loses on the track			P11	return
Two piles of 7			P12	return
The two-pile chart			P13	return
Kim's claim that the squares repeat			P14	return

First route. K–1 P1–P5; Grades 2–3 P1–P5; Grades 4–5 P1–P6. **Return visits.** K–1 P6–P9; Grades 2–3 P6–P9; Grades 4–5 P7–P14.

Key, one entry per problem

These problem files hold no keys; the week's adult guide has them.